

Swaham Pitambar Patra

Single Inheritance:

```
1 class Parent:
2     def func1(self):
3         print("Parent class function")
4
5 class Child(Parent):
6     def func2(self):
7         print("Child class function")
8
9 obj = Child()
10
11 obj.func1()
12 obj.func2()
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

PORTS

TERMINAL

```
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS> python -u "c:\Users\Swaham Patra\
Parent class function
Child class function
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS>
```

Multiple Inheritance:

```

1 class Mother:
2     def func1(self):
3         print("Mother class function")
4
5 class Father:
6     def func2(self):
7         print("Father class function")
8
9 class Child(Mother, Father):
10    def func3(self):
11        print("Child class function")
12
13 obj = Child()
14
15 obj.func1()
16 obj.func2()
17 obj.func3()

```

PROBLEMS

OUTPUT

DEBUG CONSOLE

PORTS

TERMINAL

```

PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS> python -u "c:\Users\
Mother class function
Father class function
Child class function
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS>

```

Multi-Level Inheritance:

```
1 class Grandfather:
2     def func1(self):
3         print("Grandfather class function")
4
5 class Father(Grandfather):
6     def func2(self):
7         print("Father class function")
8
9 class Child(Father):
10    def func3(self):
11        print("Child class function")
12
13 obj = Child()
14
15 obj.func1()
16 obj.func2()
17 obj.func3()
```

PROBLEMS OUTPUT DEBUG CONSOLE PORTS

TERMINAL

```
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS> python -u "c:\Users\Swaham Patra\OneDrive\ITR Python+DS\multi_level_inheritance.py"
Grandfather class function
Father class function
Child class function
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS>
```

Hierarchical Inheritance:

```
1 class Parent:
2     def func1(self):
3         print("Parent class function")
4
5 class Child1(Parent):
6     def func2(self):
7         print("Child1 class function")
8
9 class Child2(Parent):
10    def func3(self):
11        print("Child2 class function")
12
13 obj1 = Child1()
14 obj1.func1()
15 obj1.func2()
16
17 obj2 = Child2()
18 obj2.func1()
19 obj2.func3()
```

PROBLEMS OUTPUT DEBUG CONSOLE PORTS

TERMINAL

```
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS> python -u "c:\Users\Swaham Patra\OneDrive\ITR Python+DS\hierarchical_inheritance.py"
Parent class function
Child1 class function
Parent class function
Child2 class function
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS>
```

Hybrid Inheritance :

```
1 class Grandparent:
2     def func1(self):
3         print("Grandparent class function")
4
5 class Father(Grandparent):
6     def func2(self):
7         print("Father class function")
8
9 class Mother(Grandparent):
10    def func3(self):
11        print("Mother class function")
12
13 class Child(Father, Mother):
14    def func4(self):
15        print("Child class function")
16
17 obj = Child()
18 obj.func1()
19 obj.func2()
20 obj.func3()
21 obj.func4()
```

PROBLEMS OUTPUT DEBUG CONSOLE PORTS

TERMINAL

```
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS> python -u "c:\Users\Swaham Patra\OneDrive\ITR Python+DS\Tasks\07-07\inheritance\hybrid_inheritance.py"
Grandparent class function
Father class function
Mother class function
Child class function
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS>
```

Problem Statement 1:

```
1 class Person:
2     def __init__(self, name, age):
3         self.name = name
4         self.age = age
5
6 class Student(Person):
7     def __init__(self, name, age, roll_no):
8         Person.__init__(self, name, age)
9         self.roll_no = roll_no
10
11    def show(self):
12        print(f"Name: {self.name}, Age: {self.age}, Roll No.: {self.roll_no}")
13
14 obj = Student("Swaham", 18, 45)
15 obj.show()
```

PROBLEMS OUTPUT DEBUG CONSOLE PORTS

TERMINAL

```
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS> python -u "c:\Users\Swaham Patra\OneDrive\ITR Python+DS\Tasks\07-07\inheritance\problem_statement_1.py"
Name: Swaham, Age: 18, Roll No.: 45
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS>
```

Problem Statement 2:

```
1 class Vehicle:
2     def start(self):
3         print("Starting...")
4
5 class Car(Vehicle):
6     def drive(self):
7         print("Driving...")
8
9 obj = Car()
10 obj.start()
11 obj.drive()
```

PROBLEMS OUTPUT DEBUG CONSOLE PORTS

✓ TERMINAL

PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS> pyth
Starting...
Driving...
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS> █

Problem Statement 3:

```
1 class Person:
2     def __init__(self, name):
3         self.name = name
4
5 class Student(Person):
6     def __init__(self, name, roll_no):
7         Person.__init__(self, name)
8         self.roll_no = roll_no
9
10 class Employee(Person):
11     def __init__(self, name, emp_id):
12         Person.__init__(self, name)
13         self.emp_id = emp_id
14
15 class Teaching_Assistant(Student, Employee):
16     def __init__(self, name, roll_no, emp_id):
17         Student.__init__(self, name, roll_no)
18         Employee.__init__(self, name, emp_id)
19
20     def show(self):
21         print(f"Name: {self.name}, Roll No.: {self.roll_no}, EMP_ID: {self.emp_id}")
22
23 S = Teaching_Assistant("Swaham", 45, 123456789)
24 S.show()
```

PROBLEMS OUTPUT DEBUG CONSOLE PORTS

✓ TERMINAL

PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS> python -u "c:\Users\Swaham Patra\OneDrive\ITR Python+DS\Tasks 07-07\inheritance\pr
Name: Swaham, Roll No.: 45, EMP_ID: 123456789
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS> █